

SCROOGE is the ideal solution for companies wondering on how to take a step towards efficient power estimations. Its capability to provide results for a complete mixed-signal design enables precise power optimisations and sizing of the components.

Key benefits. It is functionally very easy to integrate SCROOGE in your flow as it is easy to reduce power consumption all along the design chain through accurate transient analysis of the peaks on your SoC. You can easily optimise the sequencing of an SoC by identifying the most consuming parts, without any need for any other data. SCROOGE lets you avoid design iterations with your P&R provider with a clock tree emulation available after synthesis.

SCROOGE TLA grants logic, analogue and mixed-signal design engineers with the capability to quantify power consumption, to track and locate all power peaks. As the logic part of a design is only one face of the coin, analogue designers share the need to address other issues in time domain, beginning with the resilience of switching margins. SCROOGE TLA is an EDA solution that provides the capability to analyse hierarchically the mixed-signal sensitivity to power consumption, which is a key feature for yield optimisation of logic.

Interactive analysis. SCROOGE allows concurrent simulation of mixed-signal transient analysis and power consumption along with display of analogue, leakage and dynamic power consumption as well as intrinsic capacitances for each power supply during transient simulation. It helps in the identification of critical points with current peaks in correlation with intrinsic capacitances.

How it helps in reducing power consumption. The software lets you avoid design oversizing, resulting in improved density and speed. It detects circuit defaults and predicts power consumption much earlier in the flow, thus saving designers' time by providing them with an easy, efficient and usable solution.

Reducing the peaks of a dynamic mixed-signal power consumption

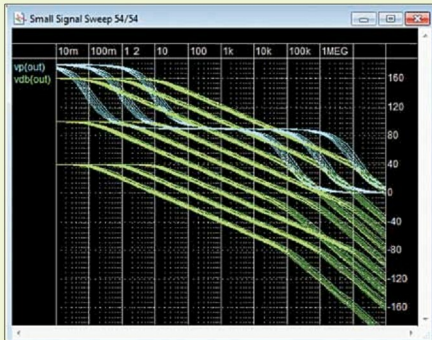


Fig. 5: Small Signal Sweep in SMASH tool

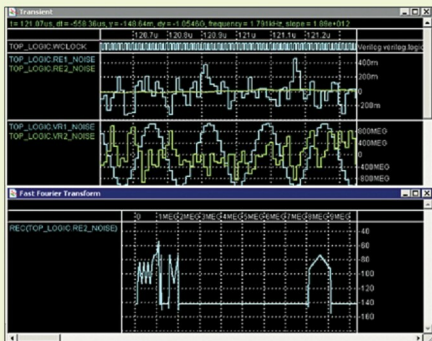


Fig. 6: Transients and FFT in SMASH tool

simulation grants the consistent benefit of smoothing-out the disturbances threatening high-resolution analogue parts, including sensitive read-margins of logic parts.

Power analysis is best performed in the time domain, while noise resilience is best analysed over the spectral domain. SCROOGE is an add-on to

all the ASIC options supporting logic structural and behavioural languages, empowering the true mixed-signal simulator with capabilities to simulate dynamic power consumption and leakage. Back-annotation of SPEF parasitic capacitances provides the ability to check with high accuracy the model for consumption after layout.